ENGINEERING DIAGRAMS · OCTOBER 2026

System design diagrams

Architecture, flowcharts, lifecycles, sequences and data model for the UK restaurant operations app, including the Maginhawa EOD trading module and its migration from the current Netlify + Power Automate + Excel setup. Companion to the Tech stack and database plan and schema.sql (v0.2).

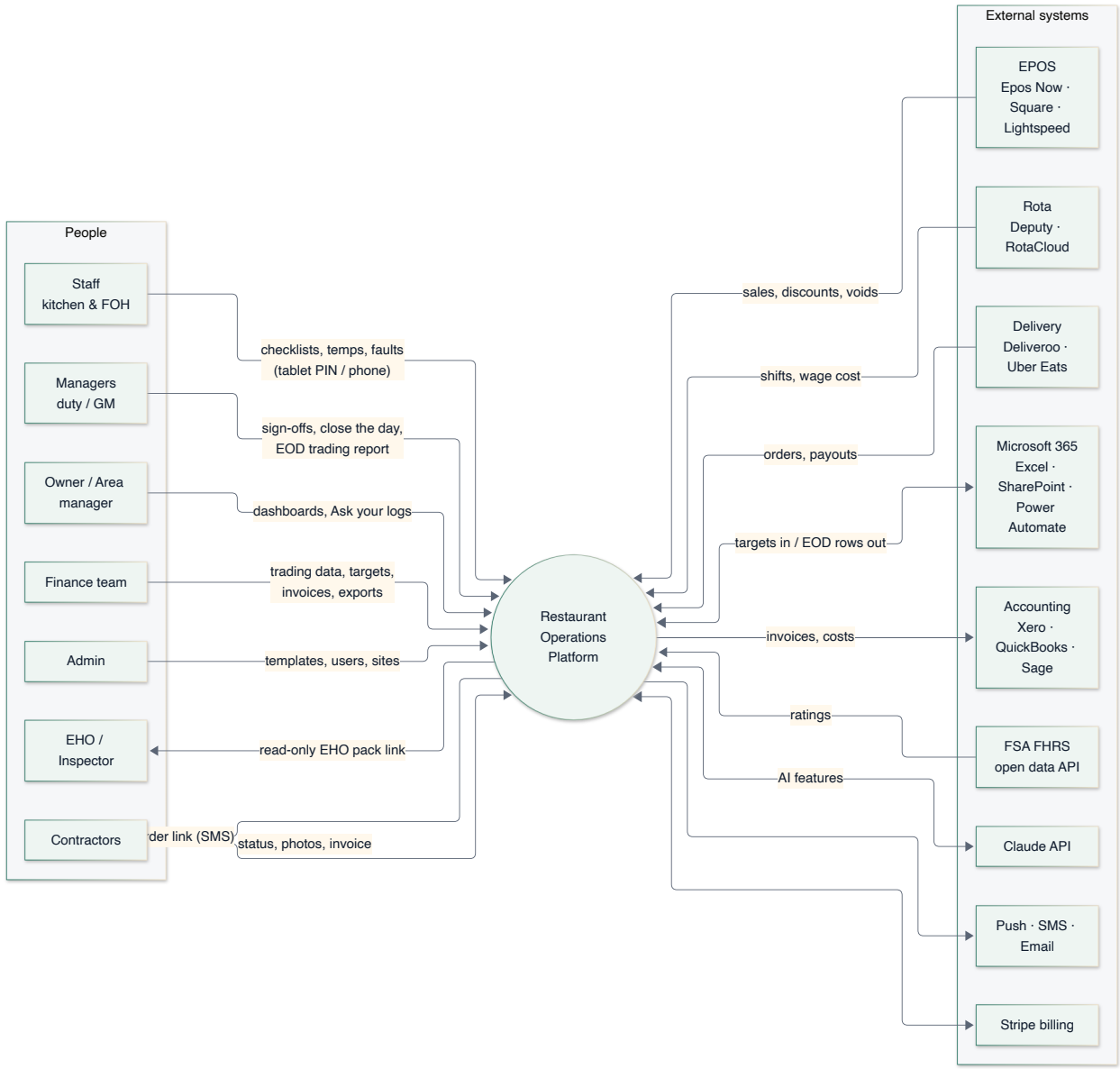
Contents

1. SYSTEM DESIGN
   System context
2. SYSTEM DESIGN
   Container architecture
3. SYSTEM DESIGN
   Who uses which device
4. FINANCE EOD: TODAY VS FUTURE
   Current Maginhawa EOD system (as-is)
5. FINANCE EOD: TODAY VS FUTURE
   Migration in three stages
6. FINANCE EOD: TODAY VS FUTURE
   Excel / SharePoint sync (Stage 1–2)
7. FLOWCHARTS
   Close-the-day flow (manager)
8. FLOWCHARTS
   Trading calculations (corrected)
9. FLOWCHARTS
   The closed loop: failed check to repair cost
10. FLOWCHARTS
    Shared tablet: PIN login
11. LIFECYCLES
    Checklist run lifecycle
12. LIFECYCLES
    Work order lifecycle
13. SEQUENCES
    Offline capture and sync
14. SEQUENCES
    Ask your logs (AI)
15. DATA MODEL
    Data model: tenancy, checklists and the compliance record
16. DATA MODEL
    Data model: repairs and maintenance
17. DATA MODEL
    Data model: trading / EOD finance module
18. DATA MODEL
    Data model: handbook, people records and allergens
19. DELIVERY PLAN
    Build roadmap

SYSTEM DESIGN · 1/19

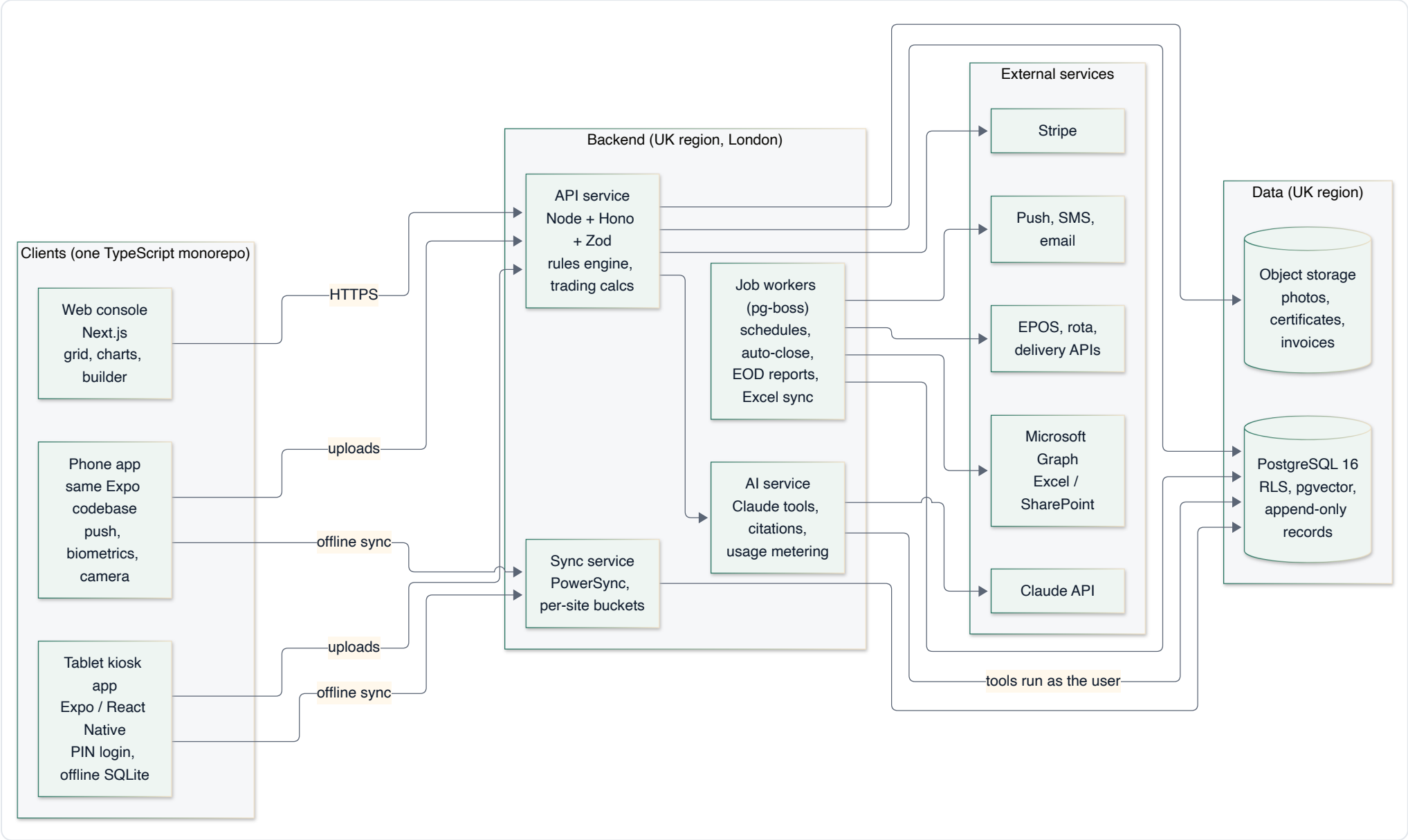
System context

Who uses the platform and which outside systems it talks to. Staff, managers, owners, finance and admins work inside it; inspectors and contractors get links; EPOS, rota, delivery, Microsoft 365, accounting, FSA, Claude, messaging and Stripe sit around it.



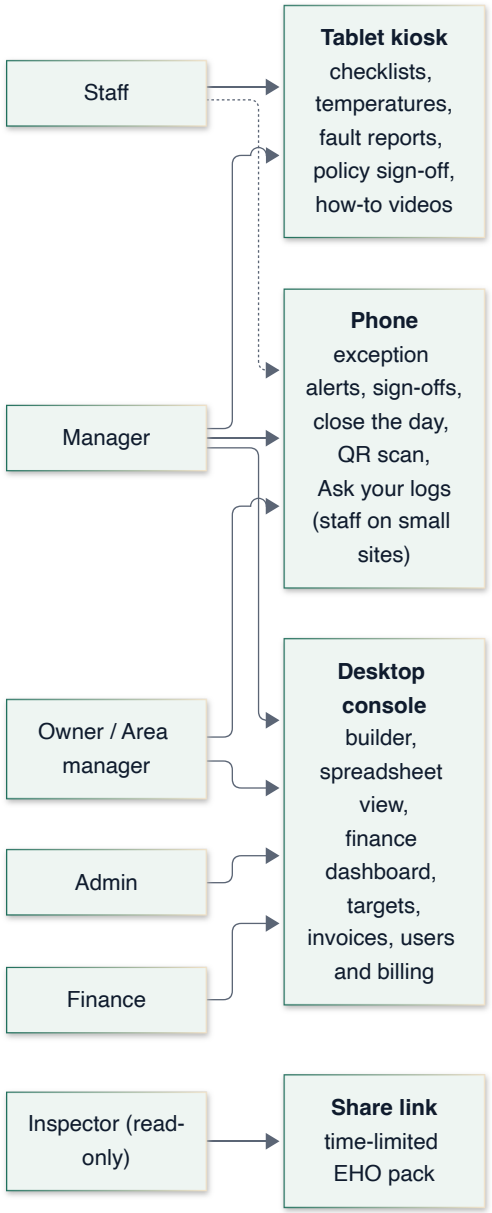
Container architecture

The deployable pieces. Tablet and phone apps sync offline through PowerSync; the web console calls the API. API, job workers, AI service and PDF renderer run in a London region on one PostgreSQL database plus object storage.



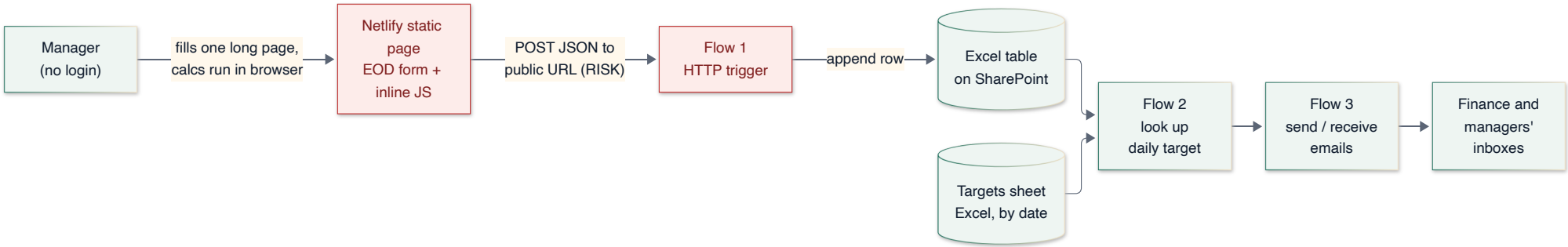
Who uses which device

Your four roles plus Owner/Area manager and a read-only inspector, mapped to the tablet kiosk, phone, desktop console and share links.



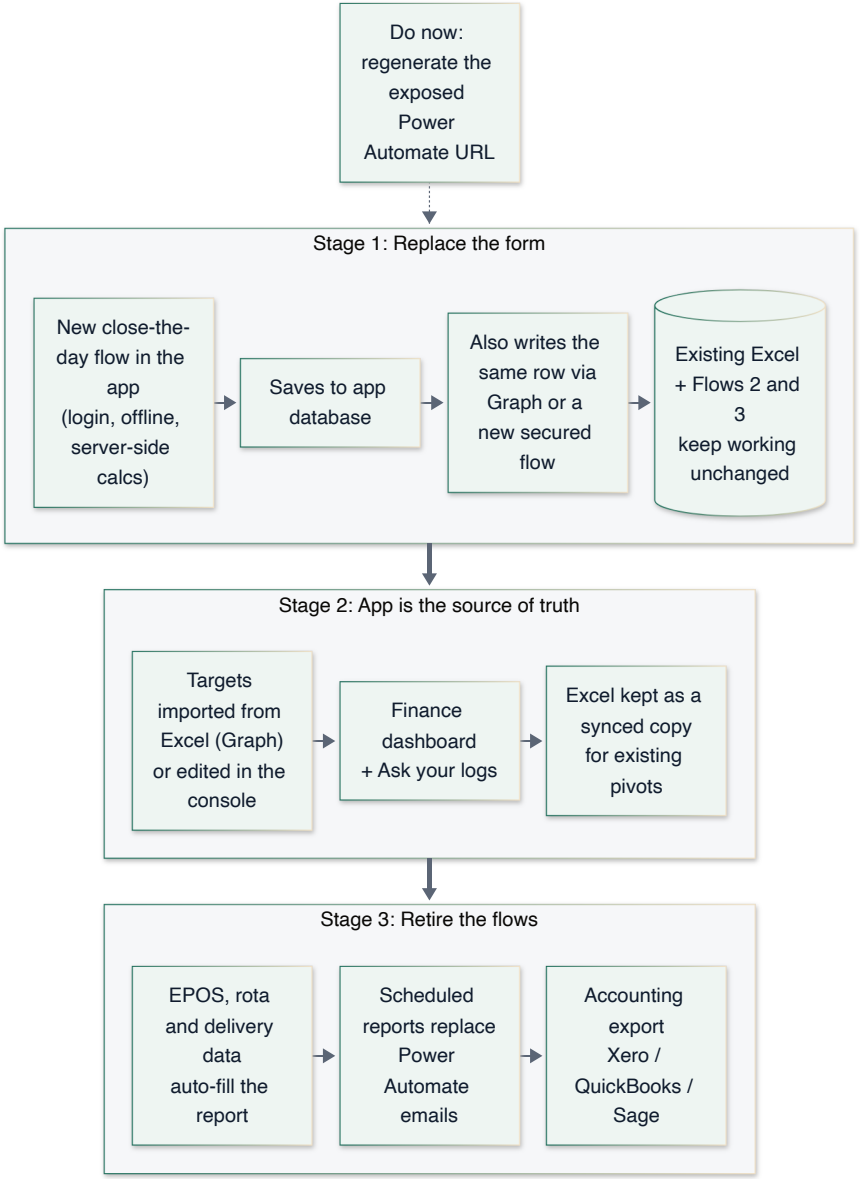
Current Maginhawa EOD system (as-is)

What the live Netlify form does today: browser-only calculations, a public Power Automate URL (red = risk), rows appended to an Excel table on SharePoint, a targets lookup and emails.



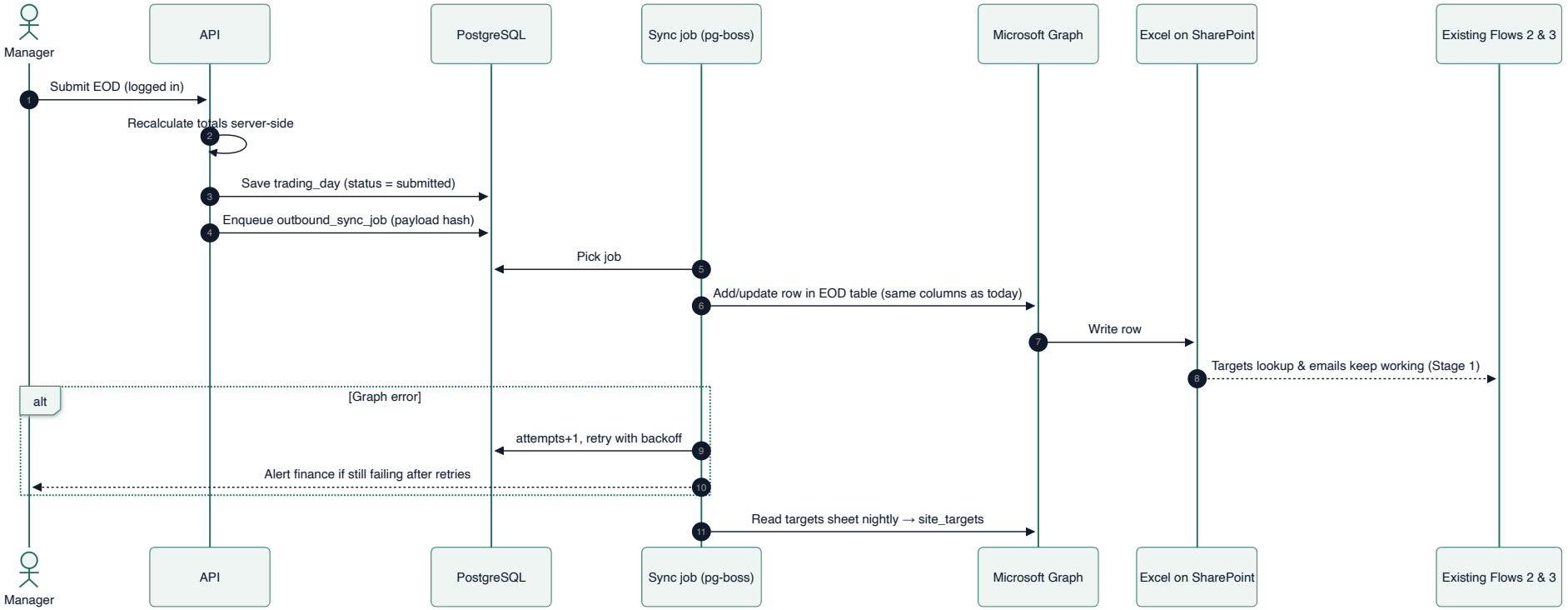
Migration in three stages

Replace the form first while keeping finance's Excel and flows working, then make the app the source of truth, then retire the flows once EPOS, rota and delivery data fill the report automatically.



Excel / SharePoint sync (Stage 1–2)

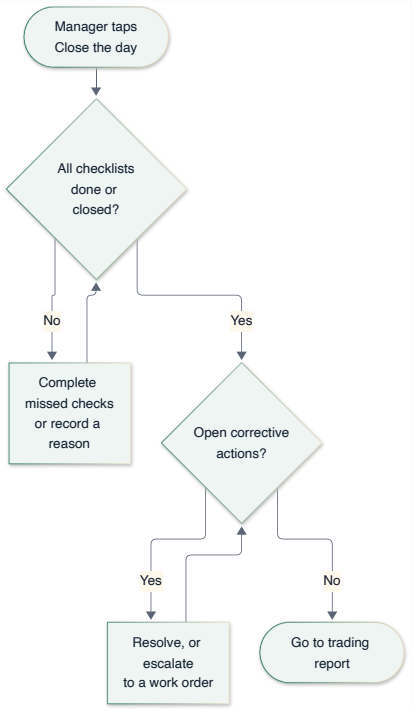
How a submitted day reaches finance's existing workbook through Microsoft Graph, idempotently and with retries, and how targets flow back in nightly.



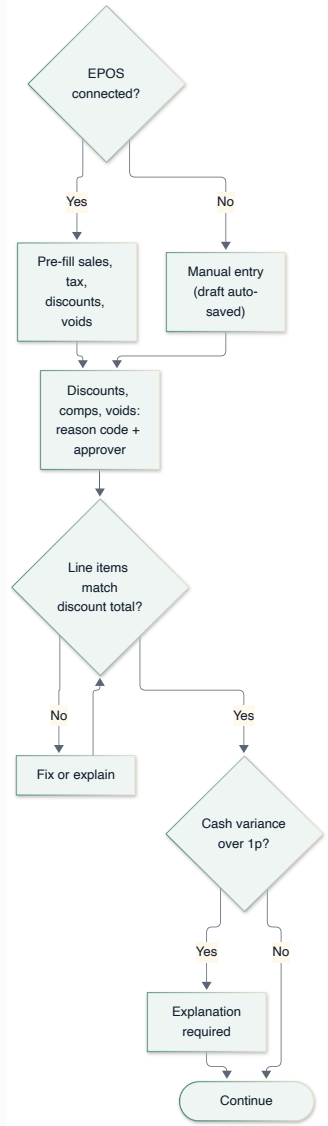
Close-the-day flow (manager)

One flow combining the ops end-of-day (checks, corrective actions) with the trading report (sales, adjustments, cash, delivery, staff, wages, feedback). Urgent maintenance becomes a work order; submission recalculates on the server, then emails, syncs to Excel and feeds dashboards.

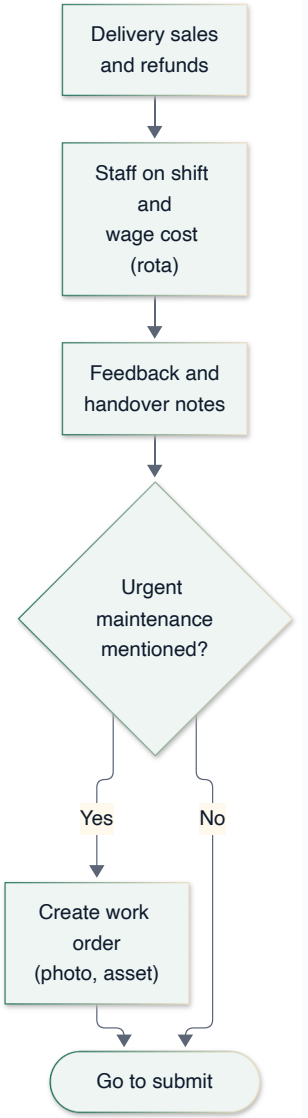
1 · Ops checks



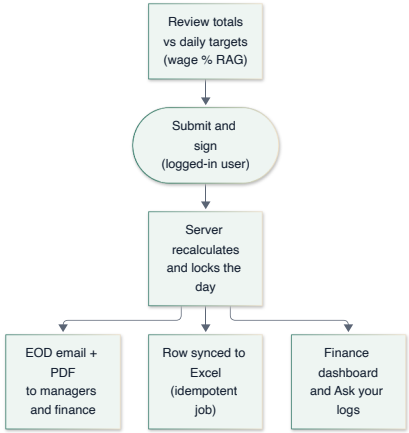
2 · Sales, adjustments, cash



3 · Delivery, staff, feedback

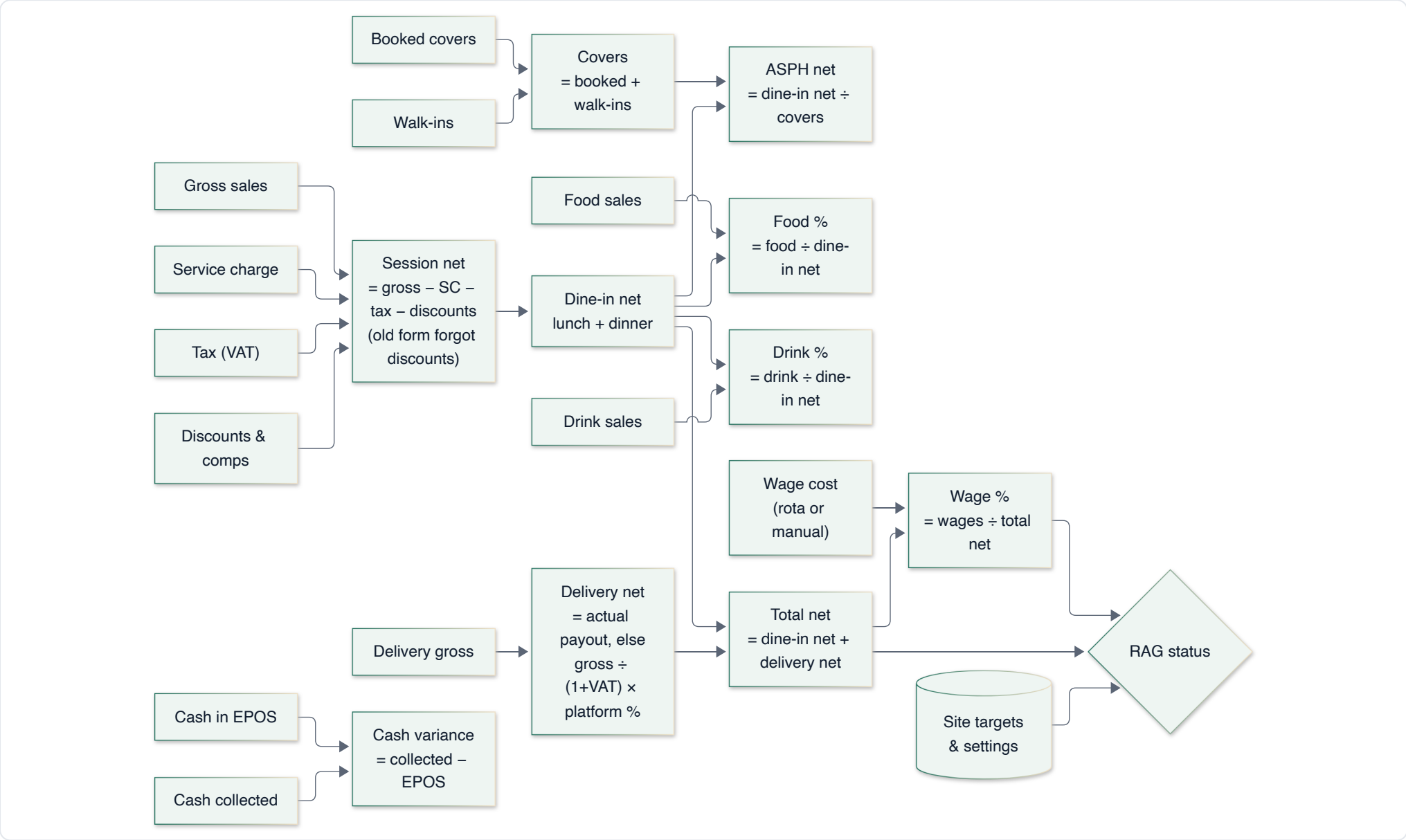


4 · Submit and outputs



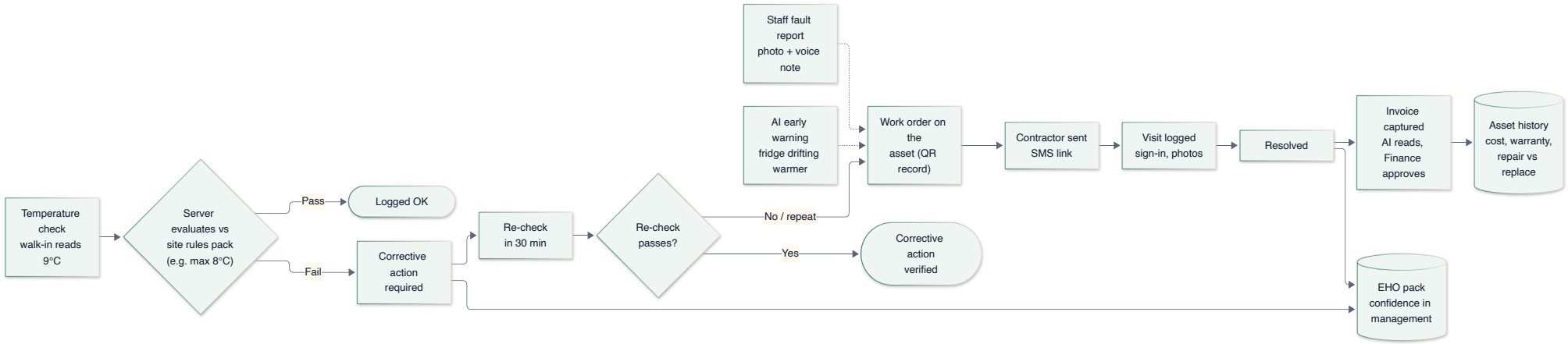
Trading calculations (corrected)

Every derived number in the EOD report and what feeds it. Fixes the old form: discounts are subtracted from net, food/drink % are labelled against net, delivery net uses real payouts or per-platform settings, and wage targets come from site settings.



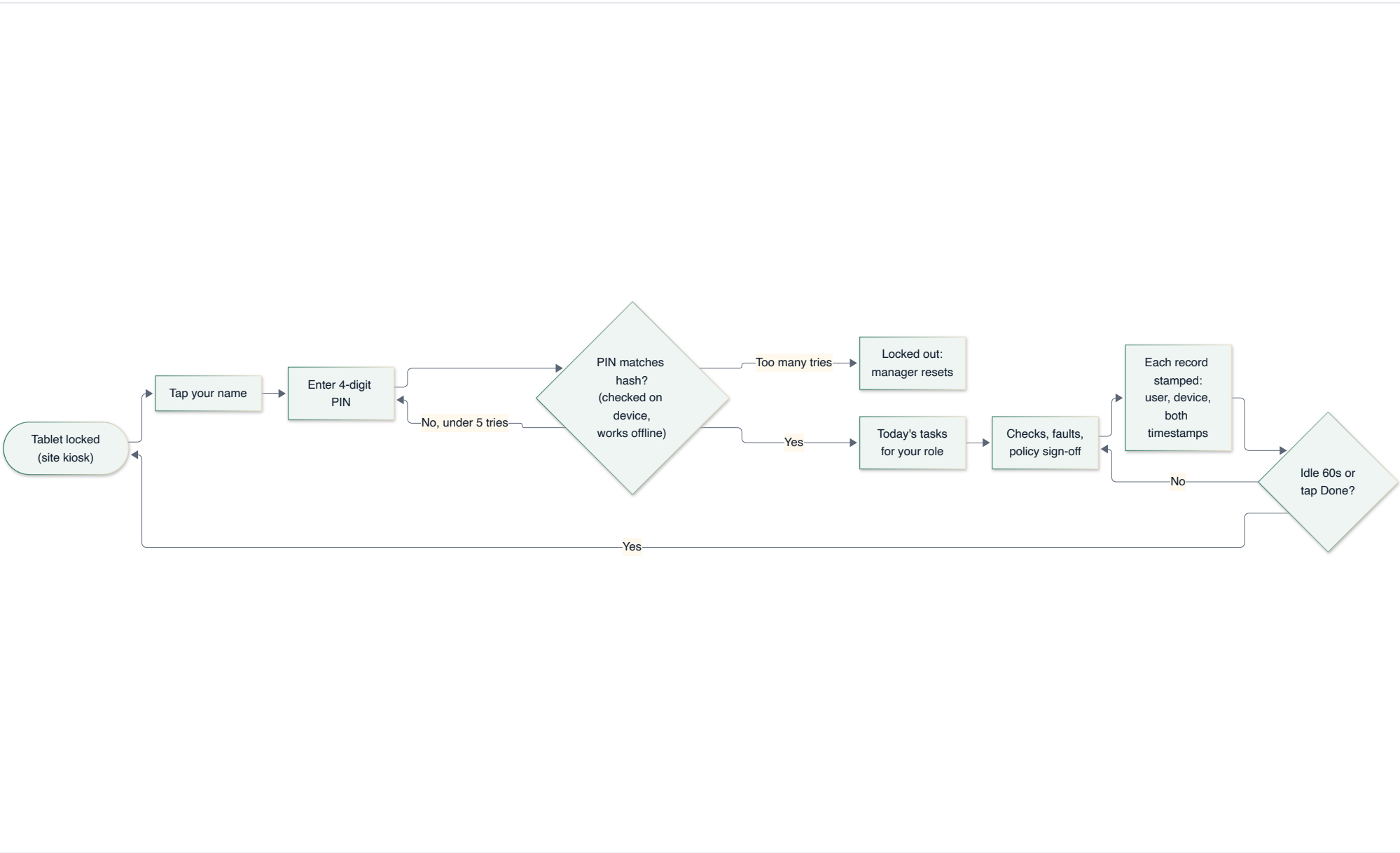
The closed loop: failed check to repair cost

The core differentiator. A failed temperature check forces a corrective action and a re-check; repeat failures, AI early warnings and staff fault reports create a work order that runs through contractor, invoice and asset cost history, and feeds the EHO pack.



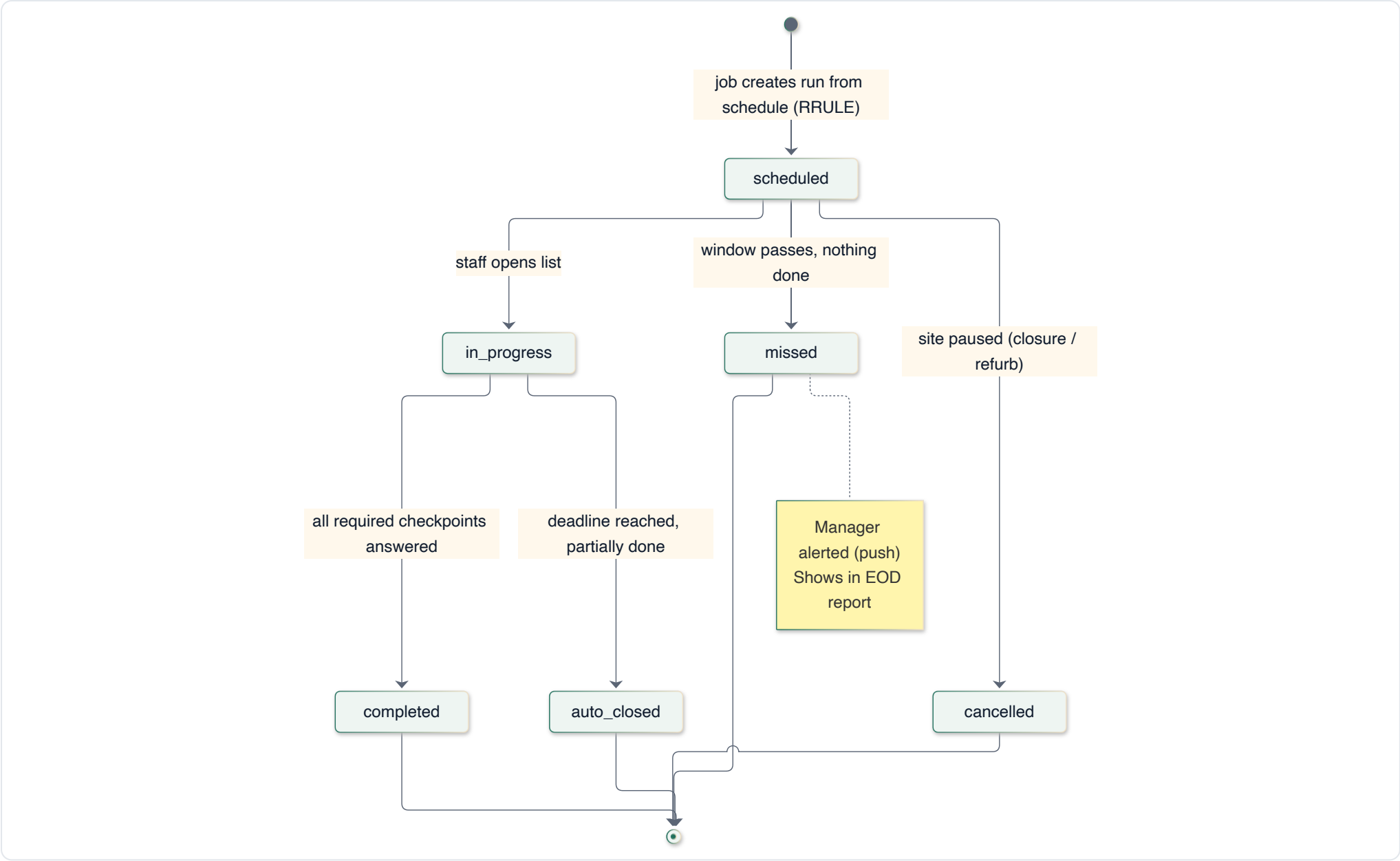
Shared tablet: PIN login

How staff use the kitchen tablet: pick name, enter PIN (verified on-device so it works offline), work, auto-lock. Every record is stamped with person, device and both timestamps.



Checklist run lifecycle

States a scheduled checklist moves through, including missed, auto-closed and cancelled (site closures).



LIFECYCLES · 12/19

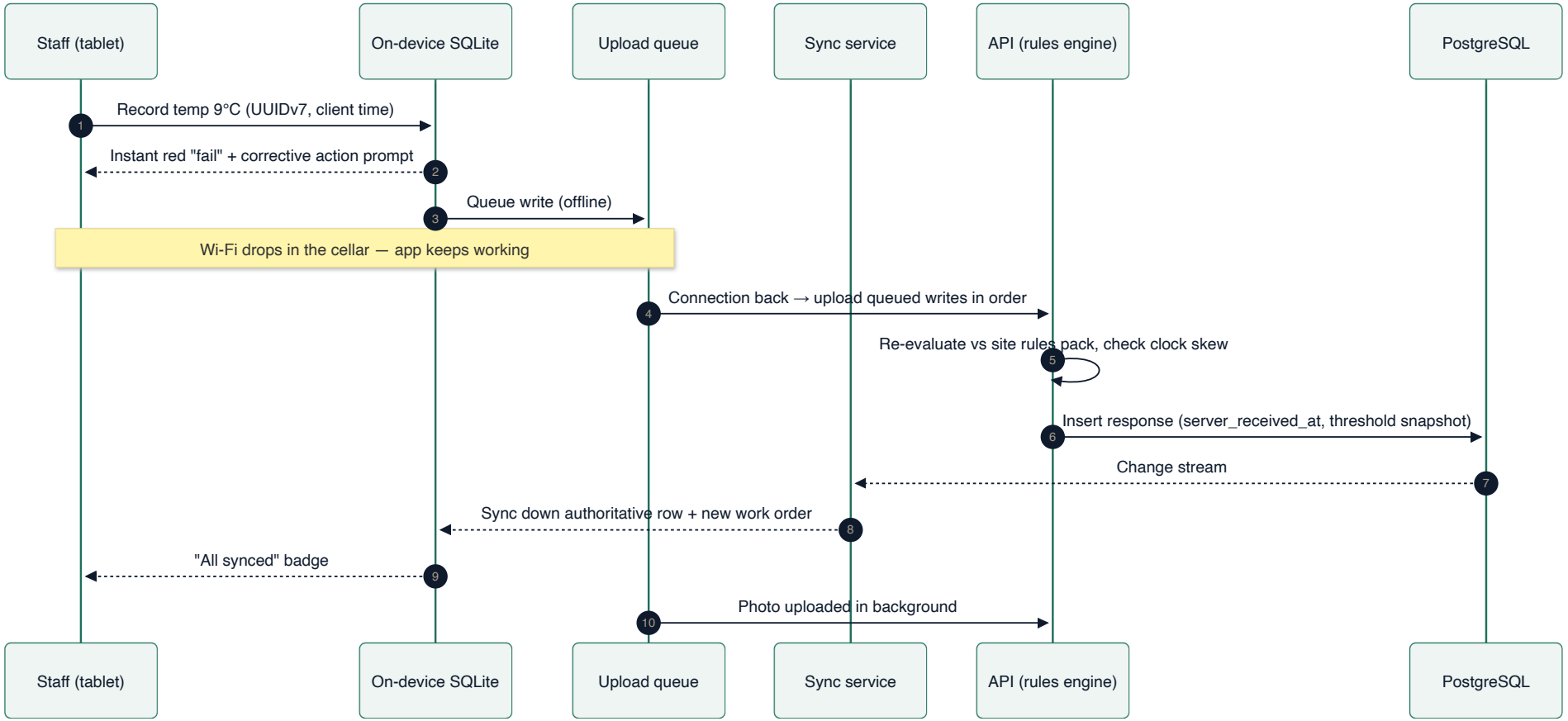
Work order lifecycle

From report to closed, including awaiting parts and cancellation; 'closed' requires the invoice to be matched and approved.



Offline capture and sync

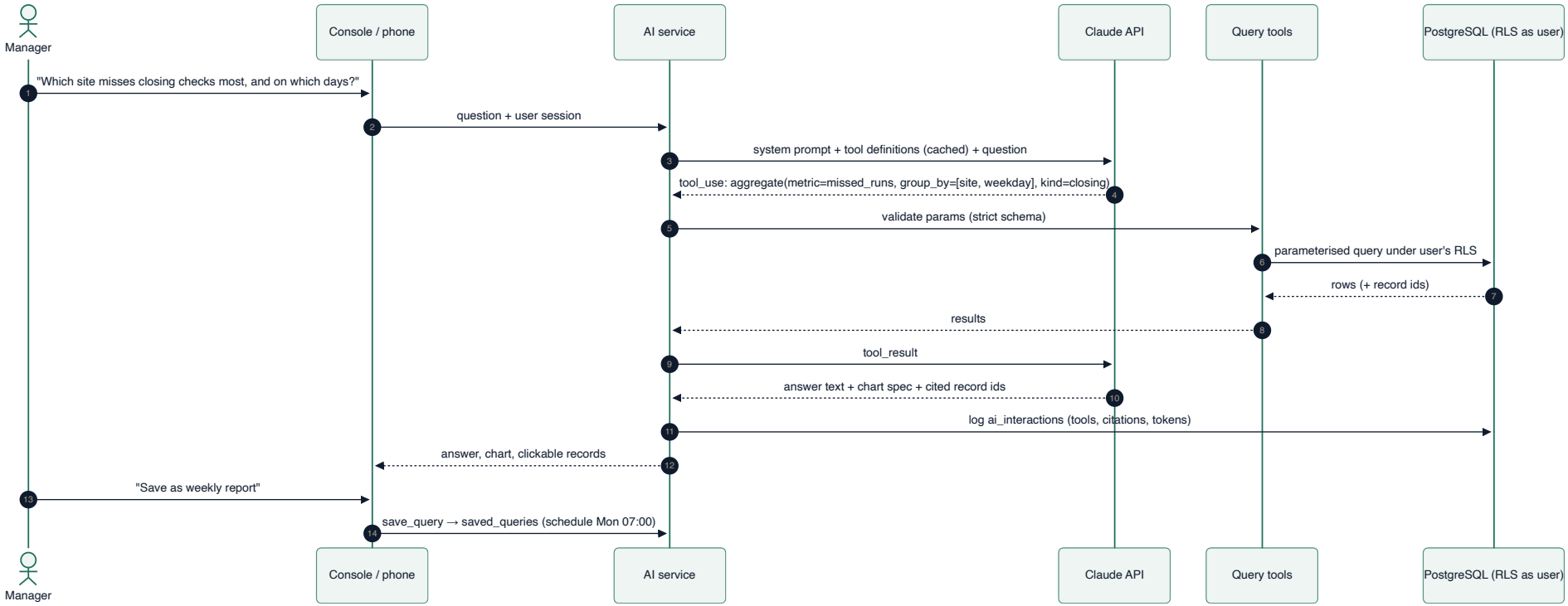
What happens when a check is recorded in a cellar with no signal: instant local feedback, queued upload, server re-evaluation, then the authoritative record (and any new work order) syncs back.



SEQUENCES · 14/19

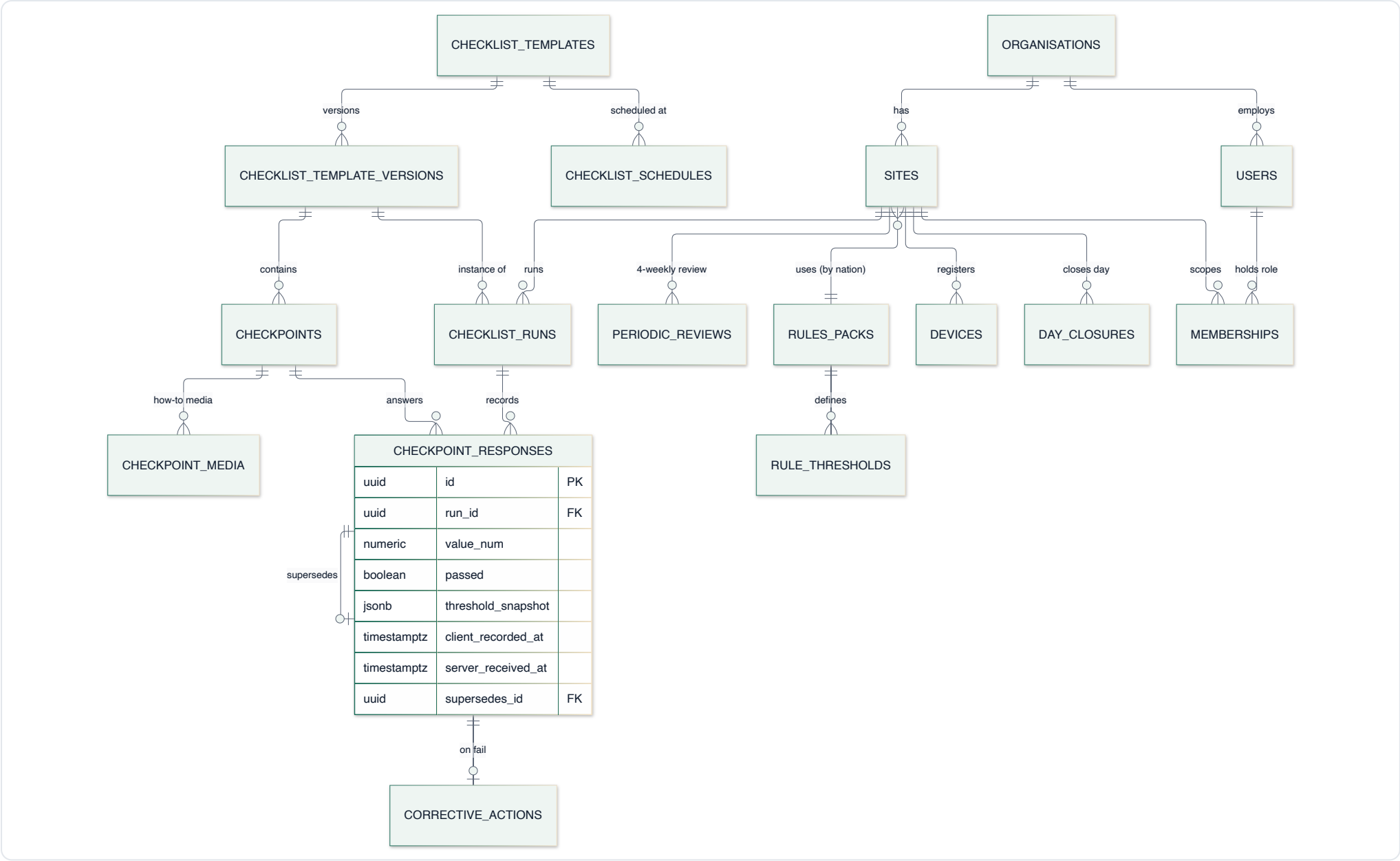
Ask your logs (AI)

A manager's question is answered by Claude choosing the app's own query tools, which run under the manager's database permissions. Results come back with chart and record citations, are logged, and can be saved as a scheduled report.



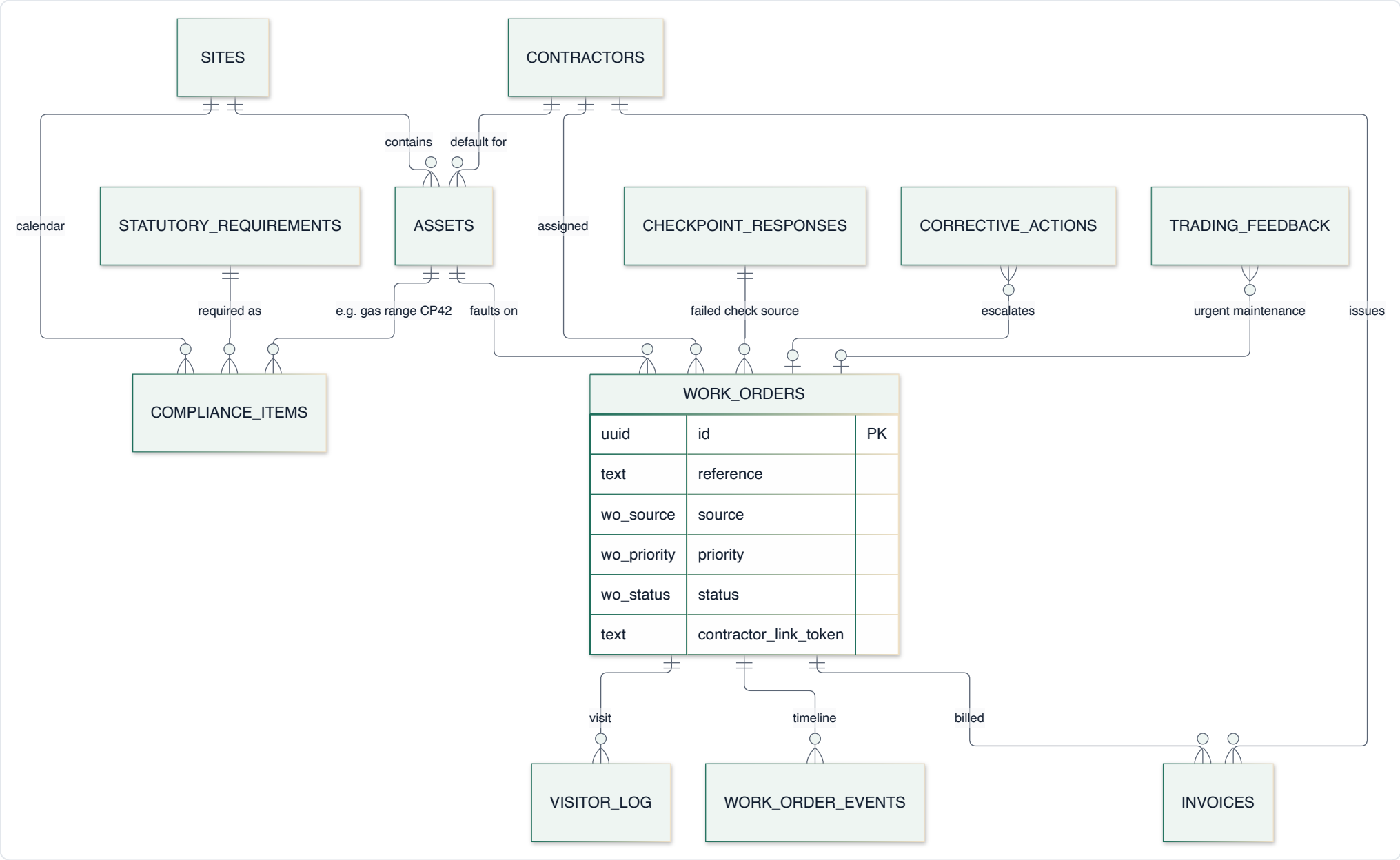
Data model: tenancy, checklists and the compliance record

Organisations, sites, users and roles; versioned checklist templates and checkpoints; append-only runs, responses and corrective actions.



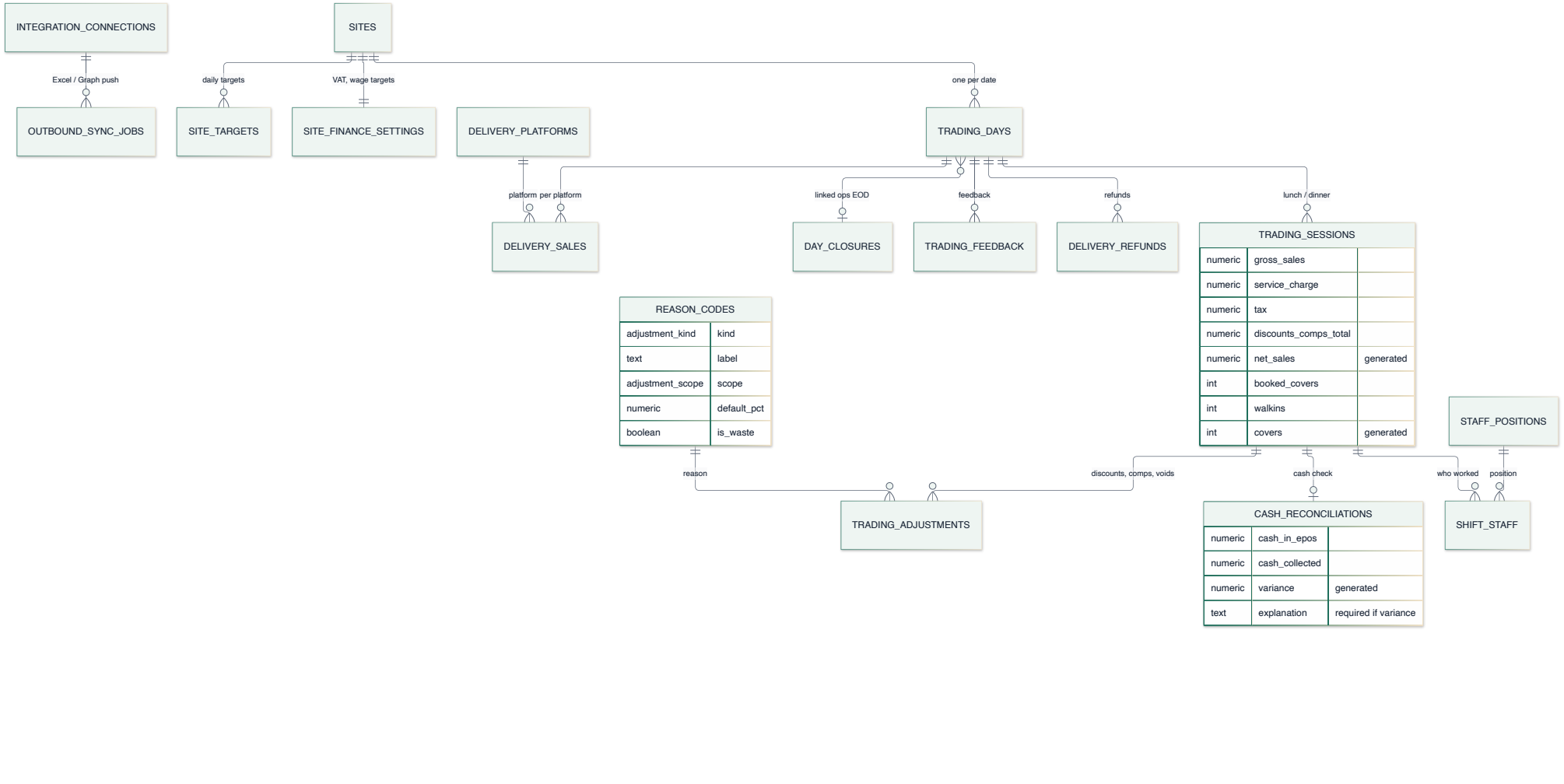
Data model: repairs and maintenance

Assets, contractors, work orders and their timeline, invoices, the statutory compliance calendar and visitor log, including links from failed checks and EOD feedback.



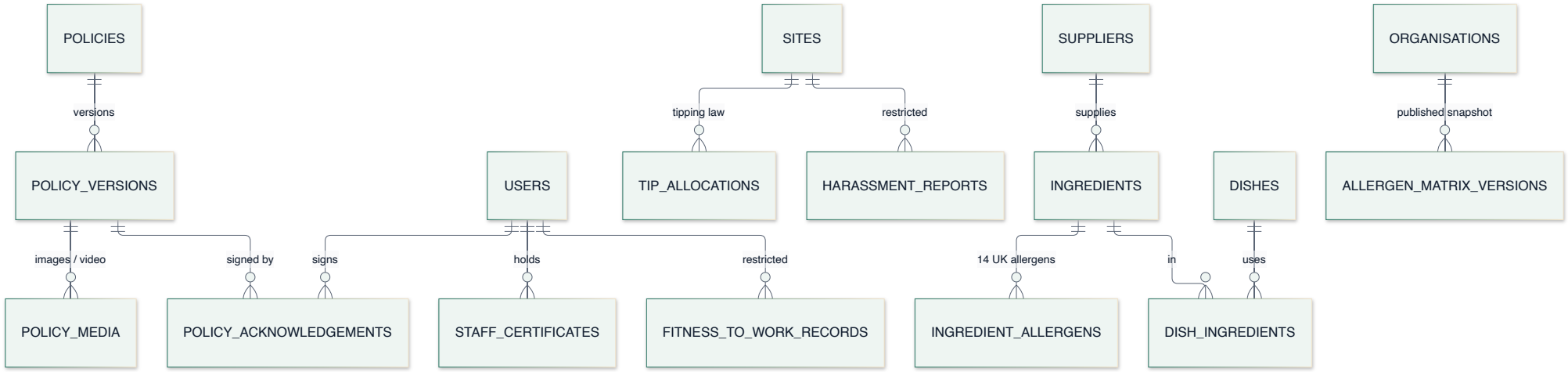
Data model: trading / EOD finance module

Replaces the Netlify form's text blobs with structured records: trading days and sessions, discounts/comps/voids with reason codes, cash reconciliation, delivery sales and refunds, staff on shift, targets, and Excel sync jobs.



Data model: handbook, people records and allergens

Versioned policies with media and e-signatures, staff certificates, restricted HR records, and the 14-allergen ingredient and dish model.



Build roadmap

Indicative sequencing from foundations through MVP, the trading module (Stage 1 and 2), AI features, a pilot at Maginhawa sites, then phase-2 integrations.

